

Diliberto–Straus Convergence Without Closedness for Two-Point Fibres

Independent research note

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Abstract

Asgarova and Ismailov ask whether their Diliberto–Straus residual algorithm converges to the uniform approximation error without assuming that the sum of the two algebras is closed. We prove a positive answer whenever every quotient fibre of both algebras has at most two points. Under the paper’s C -property, these fibres are the orbits of continuous involutions, so the midrange maps become linear. A Laurent-polynomial calculation then gives an $O(n^{-1/2})$ decay estimate on each summand. We also construct two such algebras whose sum is not closed, showing that the result genuinely passes beyond the source theorem.

Packet status

Run	fa_banach_001, lane 13.
Source	arXiv:1603.07073, Theorem 2.3 and the question on PDF page 7 [1].
Classification	Candidate strong partial result, likely valid subject to human review.
Scope	A complete answer for the two-point-fibre class; no claim for arbitrary fibres.

Thus the operator F is non-expansive. $\square$

Consider the iterations

$$h_1(x) = h(x), \quad h_{2n} = h_{2n-1} - Fh_{2n-1}, \quad h_{2n+1} = h_{2n} - Gh_{2n}, \quad n = 1, 2, \dots$$

From Theorem 2.2 and the above-mentioned general result of Golomb (see Theorem 1.1) one can obtain the following theorem.

Theorem 2.3. *Let all the assumptions of Theorem 2.2 hold and $A_1 + A_2$ be closed in $C(X)$. Then $\|h_n\|$ converges to the error of approximation $E(h)$.*

Does the Diliberto and Straus algorithm converge without the closedness assumption on the sum $A_1 + A_2$? We do not yet know a complete answer to this question. Recall that we obtain Theorem 2.3 by using the general result of Golomb (see Theorem 1.1), in which the “closedness” is a major hypothesis. Below we give a different (and independent of Golomb’s result) proof of this theorem, where it is shown how we use the mentioned closedness. It should be remarked that in the simplest case of approximation by sums of univariate functions, the known classical proofs of the Diliberto-Straus algorithm require that one could form a closed bolt (for this terminology see below) from an arbitrary bolt $(x_1, \dots, x_n)$ of a given compact set $Q \subset \mathbb{R}^2$ by adding a point $y \in Q$, whose first and second coordinates are equal to that of x_1 and x_n respectively (see, e.g., [5, 14, 17]). Note that this point may not lie in Q , unless Q is a Cartesian product of two compact sets. The main idea behind our proof is to make use of weak* cluster points of some sequence of “unclosed bolt functionals” instead of considering only “closed bolt functionals”. We hope that this idea can be useful in future attempts to prove the convergence of the Diliberto-Straus algorithm without the closedness assumption.

For the further analysis we need the following objects called “bolts of lightning” or

1 Set-up and statement

Let X be compact Hausdorff and let $\mathcal{A}_1, \mathcal{A}_2$ be closed unital real subalgebras of $C(X)$. Write $s_i : X \rightarrow X_i$ for the quotient maps induced by equality of all functions in $\mathcal{A}_i$. The source assumes the *C-property*: for every $h \in C(X)$, the maximum and minimum of h on each s_i -fibre vary continuously on X_i . Its proximity maps are the fibre midranges

$$F_i h(x) = \frac{1}{2} \left(\max_{s_i(y)=s_i(x)} h(y) + \min_{s_i(y)=s_i(x)} h(y) \right).$$

Starting with $h_1 = h$, the algorithm alternately sets

$$h_{2n} = h_{2n-1} - F_1 h_{2n-1}, \quad h_{2n+1} = h_{2n} - F_2 h_{2n}.$$

Put $E(h) = \text{dist}(h, \mathcal{A}_1 + \mathcal{A}_2)$.

Theorem 1 (two-point-fibre convergence). *Assume the C-property and*

$$\#s_i^{-1}(s_i(x)) \leq 2 \quad (x \in X, \quad i = 1, 2).$$

Then, whether or not $\mathcal{A}_1 + \mathcal{A}_2$ is closed,

$$\lim_{j \rightarrow \infty} \|h_j\|_\infty = E(h) \quad (h \in C(X)).$$

More precisely, there are continuous involutions $\sigma, \tau : X \rightarrow X$ such that $\mathcal{A}_1 = \text{Fix } S_\sigma$, $\mathcal{A}_2 = \text{Fix } S_\tau$, where $S_\sigma h = h \circ \sigma$. If

$$T = \frac{I - S_\tau}{2} \frac{I - S_\sigma}{2},$$

then $h_{2n+1} = T^n h$, and for every $v \in \mathcal{A}_2$,

$$\|T^n v\|_\infty \leq \frac{\|v\|_\infty}{\sqrt{n}} \quad (n \geq 1). \quad (1)$$

Idea of the proof

There are two rigidities hidden in the two-point hypothesis. First, the source's C -property forces the operation that swaps the two members of a fibre to be continuous. Each midrange map is therefore the average of the identity with an isometric involution. Second, alternating the complementary half-differences produces a fixed linear contraction T . On a function fixed by the second involution, every iterate of T is a Laurent polynomial in the product of the two involutions. Its coefficients are discrete derivatives of a symmetric binomial distribution and have total variation $O(n^{-1/2})$. The limiting residual norm is consequently a seminorm that vanishes on both algebras and their closure. Since every residual is in the same quotient class as the initial function, this seminorm must equal the distance to the algebraic sum.

2 Two-point fibres are involution orbits

Lemma 1. *Let $s : X \rightarrow Y$ be one of the algebra quotients, assume its fibres have at most two points, and assume the C -property. The map σ that swaps the two points of every doubleton fibre and fixes singleton fibres is a continuous involution. The associated algebra is $\text{Fix } S_\sigma$, and its midrange map is $(I + S_\sigma)/2$.*

Proof. For $f \in C(X)$, let Mf be its fibre midrange, regarded as a continuous function on X via s . On a singleton or doubleton fibre,

$$f(\sigma(x)) = 2Mf(x) - f(x).$$

Thus $f \circ \sigma$ is continuous for every $f \in C(X)$. Continuous real functions determine the topology of a compact Hausdorff space, so σ is continuous. It is visibly an involution and its orbits are exactly the fibres. By Stone–Weierstrass, the closed unital algebra is all continuous functions on its Hausdorff quotient; hence it consists exactly of the σ -invariant functions. The displayed identity also gives the formula for the midrange map. $\square$

This lemma also proves the converse description: a continuous involution has one- or two-point orbits, its invariant algebra has the C -property, and fibre maximum and minimum are

$$\frac{f + f \circ \sigma \pm |f - f \circ \sigma|}{2}.$$

3 The decay identity

Set $U = S_\sigma$, $V = S_\tau$ and

$$P = \frac{I - U}{2}, \quad Q = \frac{I - V}{2}, \quad T = QP.$$

These are contractions on $C(X)$, $U^2 = V^2 = I$, and $h_{2n+1} = T^n h$. Put

$$R = VU, \quad D = \frac{R - R^{-1}}{4}, \quad L = \frac{2I + R + R^{-1}}{4}.$$

Here R and every R^k are surjective isometries.

If $v \in \text{Fix } V$, a direct expansion gives $Tv = Dv$. The range of T lies in the anti-invariant space $\{x : Vx = -x\}$, and on that space another direct expansion gives $Tx = Lx$. Therefore

$$T^n v = L^{n-1} Dv \quad (n \geq 1, v \in \text{Fix } V). \quad (2)$$

For $m = n - 1$, the binomial theorem yields

$$L^m = \sum_{k=-m}^m p_m(k) R^k, \quad p_m(k) = 4^{-m} \binom{2m}{m+k},$$

where $p_m(k) = 0$ outside $[-m, m]$. The coefficient of R^k in $L^m D$ is

$$q_m(k) = \frac{p_m(k-1) - p_m(k+1)}{4}.$$

The sequence p_m is symmetric and unimodal, so its two tails telescope:

$$\sum_k |q_m(k)| = \frac{p_m(0) + p_m(1)}{2} \leq p_m(0) = 4^{-m} \binom{2m}{m} \leq \frac{1}{\sqrt{m+1}}. \quad (3)$$

The last elementary central-binomial bound also follows immediately by induction. Equations (2)–(3), together with $\|R^k\| = 1$, prove (1).

Proof of Theorem 1. Lemma 1 supplies U, V . Since T is linear and contractive, the decreasing limit

$$\rho(h) = \lim_{n \rightarrow \infty} \|T^n h\|_\infty$$

is a seminorm satisfying $\rho(h) \leq \|h\|_\infty$. If $u \in \mathcal{A}_1$, then $Tu = 0$; if $v \in \mathcal{A}_2$, estimate (1) gives $\rho(v) = 0$. Therefore ρ vanishes on $\mathcal{A}_1 + \mathcal{A}_2$ and, being 1-Lipschitz, on its closure. It follows that

$$\rho(h) = \rho(h - w) \leq \|h - w\|_\infty \quad (w \in \mathcal{A}_1 + \mathcal{A}_2),$$

so $\rho(h) \leq E(h)$. In the other direction, every update subtracts an element of one algebra, hence $h - T^n h \in \mathcal{A}_1 + \mathcal{A}_2$ and $E(h) \leq \|T^n h\|_\infty$. Passing to the limit proves $\rho(h) = E(h)$. The norms of the entire residual sequence are decreasing, while its odd subsequence has this limit, so the same is true of all $\|h_j\|_\infty$. $\square$

4 A genuinely nonclosed example

The closedness hypothesis is not automatic in Theorem 1. Let $N_j \rightarrow \infty$ be even integers and let X be the one-point compactification of the disjoint union of cycles $C_{N_j} = \mathbb{Z}/N_j\mathbb{Z}$. On every cycle define

$$\sigma(k) = -k, \quad \tau(k) = 1 - k,$$

and fix the point at infinity. These are continuous involutions. Let $\mathcal{A}_1 = \text{Fix } S_\sigma$ and $\mathcal{A}_2 = \text{Fix } S_\tau$.

On C_N , with $\theta = 2\pi/N$, put

$$u_N(k) = \cos(\theta k), \quad v_N(k) = -\cos(\theta(k - \frac{1}{2})).$$

Then $u_N \in \mathcal{A}_1|_{C_N}$, $v_N \in \mathcal{A}_2|_{C_N}$, and

$$\|u_N + v_N\|_\infty \leq \frac{\pi}{N}. \quad (4)$$

Define $y(\infty) = 0$ and $y|_{C_{N_j}} = u_{N_j} + v_{N_j}$. By (4), y is continuous, and truncating to finitely many cycles shows $y \in \overline{\mathcal{A}_1 + \mathcal{A}_2}$.

Suppose $y = a + b$ with $a \in \mathcal{A}_1$, $b \in \mathcal{A}_2$. Shift a constant between a and b so that both vanish at infinity. On each cycle, $a - u_N = -(b - v_N)$ is fixed by both involutions. The involutions generate the transitive dihedral action because their product is a one-step rotation, so their common fixed functions on C_N are constants. Since N is even, u_N takes the values 1 and -1 and has distance one from the constants. Hence the restriction of a to every cycle has distance one from the constants, contradicting $\|a|_{C_{N_j}}\|_\infty \rightarrow 0$. Thus $y \notin \mathcal{A}_1 + \mathcal{A}_2$, and the sum is not closed. Nevertheless Theorem 1 proves convergence for every $h \in C(X)$.

5 Boundary of the result

For fibres of size at least three, the midrange depends on the currently active maximum and minimum and is no longer a fixed average. The residual iteration is then nonlinear, so $\rho(h) = \lim \|h_{2n+1}\|$ need not be a seminorm and the Laurent cancellation above has no fixed isometry on which to act. Eight focused routes were audited: bounded larger fibres, the source's open-bolt correction, later infinite-bolt duality [2], an abstract asymptotic functional, and long finite gadgets. The first four all encounter this switching-extrema obstruction, while long two-point gadgets are ruled out by (3). No full proof or counterexample for arbitrary fibres is claimed.

The exact coefficient identity was checked for $0 \leq m \leq 100$, and all operator formulas, the rate, and the nonclosed witnesses were independently tested on finite dihedral cycles. A bounded literature audit on August 17, 2026 found no later explicit resolution of the source question or the two-point-fibre theorem. The result should be treated as a candidate new partial answer pending expert review.

References

- [1] A. Kh. Asgarova and V. E. Ismailov, *On the Diliberto–Straus algorithm for the uniform approximation by a sum of two algebras*, arXiv:1603.07073 (2016).
- [2] A. Kh. Asgarova and V. E. Ismailov, *A note on approximation by a sum of two algebras*, arXiv:2405.09815 (2024).